QUESTION ANALYSIS

# EEE329

Communication Engineering I

# Question Analysis

**Sessions 2020–21 to 2021–22**

*2 papers, 51 questions, 90 formulas*

INSIDE

• At a glance • Heat map • Priority list • 10 topic pages • Derivations • Repeats • Styles • Study order

Prepared by Nur Mohammad

October 2026

# Contents


**1 At a glance** **3**

**2 Heat map** **3**

**3 Priority list** **4**

**4 Topic pages** **4**

4.1 Signals and spectrum . . . . . 4
4.2 Noise and channel capacity . . . . . 5
4.3 Amplitude modulation . . . . . 6
4.4 Angle modulation . . . . . 8
4.5 Receivers . . . . . 9
4.6 Sampling and pulse modulation . . . . . 10
4.7 Digital modulation . . . . . 11
4.8 Multiplexing and multiple access . . . . . 12
4.9 Information theory . . . . . 13
4.10 System design . . . . . 14

**5 Derivations** **14**

**6 Repeat tracker** **14**

**7 Question style** **14**

**8 Study order** **15**




Free shared copy · Nur Mohammad

# 1. At a glance:

This analysis covers **2 papers** (2020-21, 2021-22), with 51 questions and 293 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

### The five topics with the most marks:

| Topic | Marks | Share | Questions | Top subtopics (marks) |
|---|---|---|---|---|
| Amplitude modulation | 57 | 19 % | 10 | AM modulation index (19); DSB-SC (14); VSB (13) |
| Sampling and pulse modulation | 48 | 16 % | 9 | PCM (20); Delta modulation (16); Sampling (12) |
| Multiplexing and multiple access | 46 | 16 % | 9 | TDM (19); Multiple access (16); Spread spectrum (6) |
| Digital modulation | 43 | 15 % | 7 | QAM (14); ASK, FSK and PSK (11); DPSK (10) |
| Angle modulation | 29 | 10 % | 5 | Deviation and index (15); FM and PM (5); NBFM (5) |

A formula name in bold is a key formula.

# 2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

| Topic and subtopic | 2020-21 | 2021-22 | All |
|---|---|---|---|
| **Signals and spectrum** | **6** | **13** | **19** |
| Propagation modes | · | **7** | 7 |
| EM spectrum | **6** | · | 6 |
| Distortionless transmission | · | **6** | 6 |
| **Noise and channel capacity** | **20** | **6** | **26** |
| Channel capacity | **16** | **6** | 22 |
| Thermal and transit-time noise | **4** | · | 4 |
| **Amplitude modulation** | **17** | **40** | **57** |
| AM modulation index | **4** | **15** | 19 |
| DSB-SC | · | **14** | 14 |
| VSB | **7** | **6** | 13 |
| Sidebands and power | **6** | · | 6 |
| Envelope detector | · | **5** | 5 |
| **Angle modulation** | **19** | **10** | **29** |
| Deviation and index | **10** | **5** | 15 |
| FM and PM | **5** | · | 5 |
| NBFM | · | **5** | 5 |
| Carson's rule | **4** | · | 4 |
| **Receivers** | **7** | · | **7** |
| Superheterodyne | **7** | · | 7 |
| **Sampling and pulse modulation** | **22** | **26** | **48** |
| PCM | **5** | **15** | 20 |
| Delta modulation | **10** | **6** | 16 |
| Sampling | **7** | **5** | 12 |
| **Digital modulation** | **19** | **24** | **43** |
| QAM | **7** | **7** | 14 |
| ASK, FSK and PSK | · | **11** | 11 |

| Topic and subtopic | 2020-21 | 2021-22 | All |
|---|---|---|---|
| DPSK | **4** | **6** | 10 |
| FSK | **8** | · | 8 |
| **Multiplexing and multiple access** | **28** | **18** | **46** |
| TDM | **7** | **12** | 19 |
| Multiple access | **10** | **6** | 16 |
| Spread spectrum | **6** | · | 6 |
| FDM | **5** | · | 5 |
| **Information theory** | **5** | **6** | **11** |
| Entropy | **5** | **6** | 11 |
| **System design** | · | **7** | **7** |
| Link design | · | **7** | 7 |

# 3. Priority list:

**Must know:** Amplitude modulation (57), Sampling and pulse modulation (48), Multiplexing and multiple access (46), Digital modulation (43), Angle modulation (29), Noise and channel capacity (26), Signals and spectrum (19), Information theory (11)

**Likely:** Receivers (7), System design (7)

**Less often:** none

**The rule:** Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

# 4. Topic pages:

## 4.1. Signals and spectrum:

19 marks, 3 questions, in 2 of 2 papers. Book: Lathi; Forouzan Ch. 3.

Most often theory (2 of 3). Subtopics: EM spectrum; Propagation modes; Distortionless transmission.

**Formulas:**

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Propagation modes | Radio horizon (LOS) | $d \approx 4.12\left(\sqrt{h_t}+\sqrt{h_r}\right)$ km | 21-22 1(b) |
| Distortionless transmission | Distortionless output | $y(t) = k\,x(t-t_d)$ | 21-22 1(d) |
| | **Distortionless channel** | $H(f) = k\,e^{-j2\pi f t_d}$ | 21-22 1(d) |
| | Group delay | $t_g = -\frac{1}{2\pi}\frac{d\theta_h}{df} = t_d$ | 21-22 1(d) |

**Questions:**

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **EM spectrum** | | | | |
| 1 | 20-21 1(b) | 6 | theory | Briefly explain the three major bands of the electromagnetic (EM) spectrum-radio waves, microwaves and infra-red waves-in terms of their bandwidth, propagation mode and applications. |
| **Propagation modes** | | | | |
| 2 | 21-22 1(b) | 7 | theory | Discuss the different modes of propagation of EM wave. |


Free shared copy · Nur Mohammad

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Distortionless transmission** | | | | |
| 3 | 21-22 1(d) | 6 | derivation | Explain the conditions required for distortionless transmission. |

## 4.2. Noise and channel capacity:

26 marks, 4 questions, in 2 of 2 papers. Book: Lathi; Forouzan Ch. 3.

Most often numerical (3 of 4). Subtopics: Thermal and transit-time noise; Channel capacity.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Thermal and transit-time noise | **Thermal noise power** | $P_n = kTB$ | 20-21 1(c) |
| | Thermal noise voltage | $V_n = \sqrt{4kTBR}$ | 20-21 1(c) |
| Channel capacity | Equally likely levels | $H = \log_2 L$ | 20-21 3(d) |
| | **Information rate** | $R = f_s H$ bit/s | 20-21 3(d) |
| | **Shannon capacity** | $C = B \log_2 \left(1 + \frac{S}{N}\right)$ | 20-21 3(d) |
| | Decibels to ratio | $\frac{S}{N} = 10^{\text{dB}/10}$ | 20-21 5(b) |
| | Minimum SNR | $\text{SNR}_{min} = 2^{R/B} - 1$ | 20-21 5(b) |
| | Bandwidth for a rate | $B = \frac{R}{\log_2(1 + S/N)}$ | 20-21 3(d) |
| | Attenuation | $10 \log_{10} \frac{P_t}{P_r} = 20 \log_{10} \frac{V_t}{V_r}$ dB | 21-22 1(c) |
| | Signal-to-noise ratio | $\frac{S}{N} = \left(\frac{V_s}{V_n}\right)^2$ | 21-22 1(c) |
| | **Shannon capacity** | $C = B \log_2 \left(1 + \frac{S}{N}\right)$ bit/s | 21-22 1(c) |
| | SNR for a rate | $\frac{S}{N} = 2^{R/B} - 1$ | 21-22 1(c) |

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Thermal and transit-time noise** | | | | |
| 1 | 20-21 1(c) | 4 | theory | Write short note on- (i) Thermal Noise (ii) Transit Time Noise |


Free shared copy · Nur Mohammad

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Channel capacity** | | | | |
| 2 | 20-21 3(d) | 10 | numerical | An analog signal having 4 kHz bandwidth is sampled at 1.25 times the Nyquist rate, and each sample is quantized into one of 256 equally likely level. Assuming the samples to be statistically independent: (i) What is the information rate of the source? (ii) Can the output of this source be transmitted without error over the AWGN channel with a bandwidth of 10 kHz and S/N ratio of 20 dB? (iii) Find S/N ratio required for error-free transmission of part ii. (iv) Find the bandwidth required for an AWGN channel for an error free transmission of output of this source if S/N ratio is 20 dB |
| 3 | 20-21 5(b) | 6 | numerical | A voice grade telephone channel has a bandwidth of 3400 Hz. If the signal to noise ratio (SNR) on the channel is 30 dB, determine the capacity of the channel. If the above channel is to be used to transmit 4.8 kbps of data, determine the minimum SNR required on the channel. |
| 4 | 21-22 1(c) | 6 | numerical | We have a communication channel with a bandwidth of 5 MHz. A signal with an amplitude of 10 V is transmitted, and the receiver detects a signal with an amplitude of 2 V and noise with an amplitude of 5 mV. (i) Calculate the attenuation. (ii) Determine the maximum channel capacity. (iii) To achieve a data rate of 100 Mbit $s^{-1}$, find the required signal level. |

## 4.3. Amplitude modulation:

57 marks, 10 questions, in 2 of 2 papers. Book: Lathi.

Most often theory (4 of 10). Subtopics: AM modulation index; Sidebands and power; VSB; DSB-SC; Envelope detector.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| AM modulation index | AM wave | $s(t) = [E_c + E_m \sin \omega_m t] \sin \omega_c t$ | 20-21 1(d) |
| | **Modulation index** | $\mu = \dfrac{E_m}{E_c}$ | 20-21 1(d) |
| | Side frequencies | $f_c \pm f_m$, amplitude $\dfrac{\mu E_c}{2}$ | 20-21 1(d) |
| | AM wave | $\varphi_{AM}(t) = [A + m(t)] \cos \omega_c t$ | 21-22 3(d) |
| | **Modulation index** | $\mu = \dfrac{m_p}{A}$ | 21-22 3(d) |
| | **AM antenna current** | $I_t = I_c \sqrt{1 + \dfrac{\mu^2}{2}}$ | 21-22 2(a) |
| Sidebands and power | Carrier power | $P_c = \dfrac{E_c^2}{2R}$ | 20-21 1(e) |
| | **Total power** | $P_t = P_c \left(1 + \dfrac{\mu^2}{2}\right)$ | 20-21 1(e), 21-22 2(a) |
| VSB | **VSB receiver filter** | $H_o(f) = \dfrac{1}{H_i(f + f_c) + H_i(f - f_c)}$ | 20-21 4(b), 21-22 2(d) |
| | VSB signal | $\Phi_{VSB}(f) = [M(f + f_c) + M(f - f_c)]\, H_i(f)$ | 21-22 2(d) |
| | **VSB filter condition** | $H_i(f + f_c) + H_i(f - f_c) = \text{constant},\ \lvert f \rvert \le B$ | 21-22 2(d) |

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| DSB-SC | DSB-SC spectrum | $S(f) = \frac{A_c}{2}[M(f-f_c) + M(f+f_c)]$ | 20-21 5(d) |
| | Coherent detector product | $V(f) = \frac{A_c}{2}M(f) + \frac{A_c}{4}[M(f-2f_c) + M(f+2f_c)]$ | 20-21 5(d) |
| | **Lowest carrier** | $f_c \geq W$ | 20-21 5(d) |
| | **Coherent detection with errors** | $e_o(t) = m(t)\cos(\Delta\omega\, t + \theta)$ | 21-22 2(b) |
| | Costas I and Q outputs | $\frac{1}{2}m(t)\cos\theta,\ \frac{1}{2}m(t)\sin\theta$ | 21-22 2(c) |
| | Costas control signal | $\frac{1}{8}m^2(t)\sin 2\theta$ | 21-22 2(c) |
| Envelope detector | Envelope detector RC | $\frac{1}{\omega_c} \ll RC < \frac{1}{2\pi B}$ | 21-22 3(b) |

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **AM modulation index** | | | | |
| 1 | 20-21 1(d) | 4 | numerical | A carrier wave of frequency 10 MHz and peak value of 10 V is amplitude modulated by a 5 kHz sine wave of amplitude 6 V. Determine the modulation index, side frequencies and amplitude of side frequencies. |
| 2 | 21-22 2(a) | 5 | numerical | The antenna current of an AM transmitter is 8 A when only the carrier is sent. But it increases to 8.93 A when the carrier is modulated by a single sine wave. Find the percentage modulation. Determine the antenna current when the percentage of modulation changes to 0.8 |
| 3 | 21-22 3(d) | 10 | diagram | For the AM signal with $m(t)$ in Fig. 1 given below, sketch $\Phi_A M(t)$ for $\mu = 0.5, \mu = 1$, and $\mu = 1.5$. |
| **Sidebands and power** | | | | |
| 4 | 20-21 1(e) | 6 | numerical | An audio frequency signal $10\sin(2\pi 500t)$ is used to amplitude modulated a carrier of $50\sin(2\pi 105t)$ *[sic: should be* $2\pi 10^5 t$*]*. Calculate: (i) Modulation index (ii) Side-band frequencies (iii) Total power delivered to the load of 600 Ω |
| **VSB** | | | | |
| 5 | 20-21 4(b) | 7 | derivation | Explain how the message signal is reconstructed from VSB signal. |
| 6 | 21-22 2(d) | 6 | derivation | Find the condition of sideband shaping filter for VSB modulation. |
| **DSB-SC** | | | | |
| 7 | 20-21 5(d) | 0 | theory | Consider a message signal m(t) with the spectrum shown in Fig. 1. The message bandwidth W = 1 kHz. This signal is applied to a product modulator, together with a carrier wave $A_c\cos(2\pi f_c t)$, producing the DSBSC modulated wave s(t). This modulated wave is next applied to a coherent detector. Assuming perfect synchronism between the carrier waves in the modulator and detector, determine the spectrum of the detector output when: (i) the carrier frequency $f_c = 1.25$ kHz and (ii) the carrier frequency $f_c = 0.75$ kHz. |
| 8 | 21-22 2(b) | 6 | theory | Explain how phase and frequency error affect the coherent detection of AM. |
| 9 | 21-22 2(c) | 8 | theory | With appropriate block diagram, briefly explain the working principle of Costas DSB-SC receiver. |
| **Envelope detector** | | | | |
| 10 | 21-22 3(b) | 5 | theory | Write a short note on the linear diode detection (envelope detector) method with proper diagrams and leveling. |


Free shared copy · Nur Mohammad

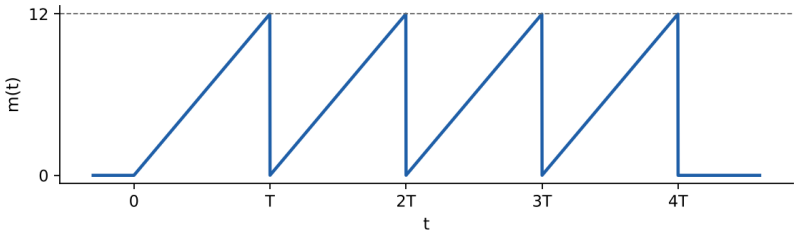



**Figure 1:** 21-22 3(d): Fig. 1 of the paper (redrawn): the message m(t)

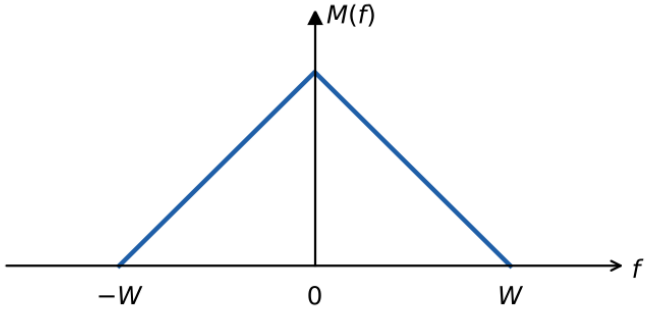



**Figure 2:** 20-21 5(d): Fig. 1 of the paper (redrawn)

## 4.4. Angle modulation:

29 marks, 5 questions, in 2 of 2 papers. Book: Lathi.

Most often theory (3 of 5). Subtopics: FM and PM; Deviation and index; Carson's rule; NBFM.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| FM and PM | Instantaneous frequency | $\omega_i = \dfrac{d\theta}{dt}$ | 20-21 2(c) |
| | PM | $\varphi_{PM} = A\cos[\omega_c t + k_p m(t)]$ | 20-21 2(c) |
| | FM | $\varphi_{FM} = A\cos\left[\omega_c t + k_f \int m\, d\alpha\right]$ | 20-21 2(c) |
| Deviation and index | **Frequency deviation** | $\Delta f = k_f V_m$ | 20-21 2(d) |
| | **FM modulation index** | $\beta = \dfrac{\Delta f}{f_m}$ | 20-21 2(d), 21-22 3(c) |
| | Frequency limits | $f_{max,min} = f_c \pm \Delta f$ | 21-22 3(c) |
| | Carrier swing | $\text{swing} = 2\Delta f$ | 21-22 3(c) |
| | Carson's rule | $B_T \approx 2(\Delta f + f_m)$ | 21-22 3(c) |
| Carson's rule | **Carson's rule** | $B_T \approx 2(\Delta f + f_m) = 2(\beta + 1) f_m$ | 20-21 4(d) |
| NBFM | FM wave | $\varphi_{FM}(t) = A\cos[\omega_c t + k_f a(t)],\ a(t) = \int m(\tau)\, d\tau$ | 21-22 3(a) |
| | **NBFM** | $\varphi_{NBFM}(t) \approx A\cos\omega_c t - A\, k_f\, a(t)\sin\omega_c t$ | 21-22 3(a) |

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **FM and PM** | | | | |
| 1 | 20-21 2(c) | 5 | theory | With proper block diagram explain the relation between frequency modulation and phase modulation. |
| **Deviation and index** | | | | |
| 2 | 20-21 2(d) | 10 | numerical | In an FM system, when the audio frequency is 500 Hz and modulating voltage 2.5 V, the deviation produced is 5 kHz. If the modulating voltage is now increased to 7.5 V, calculate the new value of frequency deviation produced. If the AF voltage is raised to 10 V while the modulating frequency dropped to 250 Hz, what is the frequency deviation? Calculate the modulation index in each case. |
| 3 | 21-22 3(c) | 5 | numerical | A 93.2 MHz carrier is frequency modulated by a 5 kHz sine wave. The resultant FM signal has a frequency deviation of 40 kHz. (i) Find the carrier swing of the FM signal. (ii) What are the highest and lowest frequencies attained by the frequency modulated signal? (iii) Calculate the modulation index for the wave. |
| **Carson's rule** | | | | |
| 4 | 20-21 4(d) | 4 | theory | State Carson's rule and its limitations. |
| **NBFM** | | | | |
| 5 | 21-22 3(a) | 5 | theory | Explain NBFM generation process with necessary block diagrams and equations. |

## 4.5. Receivers:

7 marks, 1 question, in 1 of 2 papers. Book: Lathi.

Most often theory (1 of 1). Subtopics: Superheterodyne.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Superheterodyne | **Local oscillator** | $f_{LO} = f_s + f_{IF}$ | 20-21 4(a) |
| | **Image frequency** | $f_{si} = f_s + 2f_{IF}$ | 20-21 4(a) |

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Superheterodyne** | | | | |
| 1 | 20-21 4(a) | 7 | theory | With appropriate block diagram, describe the operation of superheterodyne receiver. |


Free shared copy · Nur Mohammad

## 4.6. Sampling and pulse modulation:

48 marks, 9 questions, in 2 of 2 papers. Book: Lathi.

Most often theory (5 of 9). Subtopics: Delta modulation; PCM; Sampling.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Delta modulation | **DM largest slope** | $\sigma f_s$ | 20-21 3(a) |
| | No slope overload | $\left\lvert \frac{dm}{dt} \right\rvert_{max} \leq \sigma f_s$ | 20-21 3(a) |
| PCM | Bits per sample | $n = \log_2 L$ | 20-21 3(c) |
| | **PCM bit rate** | $R_b = n f_s$ | 20-21 3(c), 21-22 4(e) |
| | **Information rate** | $R = r\,H \quad \text{bits/s}$ | 21-22 4(c) |
| | Step size | $\Delta = \frac{2m_p}{L}$ | 21-22 4(e) |
| | Bits per sample | $L = 2^n,\ n = \log_2 L$ | 21-22 4(e) |
| | Quantization noise | $N_q = \frac{\Delta^2}{12} = \frac{m_p^2}{3L^2}$ | 21-22 4(a) |
| | **SQNR** | $\text{SQNR} = 3L^2 \frac{\overline{m^2}}{m_p^2}$ | 21-22 4(a) |
| | Peak signal to peak noise | $\frac{S_{peak}}{N_{peak}} = L^2$ | 21-22 4(e) |
| | **Minimum bandwidth** | $B_T = \frac{R_b}{2} = nB$ | 21-22 4(e) |
| | μ-law compressor | $y = \frac{\ln(1+\mu\lvert m\rvert/m_p)}{\ln(1+\mu)}\,\text{sgn}(m)$ | 21-22 4(c) |
| | Companded SQNR | $\text{SQNR} \approx \frac{3L^2}{[\ln(1+\mu)]^2}$ | 21-22 4(c) |
| Sampling | **Nyquist rate** | $f_s \geq 2B$ | 20-21 4(c), 21-22 4(b) |
| | Sampled spectrum | $\bar{G}(f) = \frac{1}{T_s}\sum_n G(f - nf_s)$ | 20-21 4(c) |
| | **Interpolation formula** | $g(t) = \sum_k g(kT_s)\,\text{sinc}(2\pi Bt - k\pi)$ | 20-21 4(c), 21-22 4(b) |
| | Sampled spectrum | $\bar{G}(f) = \frac{1}{T_s}\sum_{n=-\infty}^{\infty} G(f - nf_s)$ | 21-22 4(b) |
| | Reconstruction LPF | $H(f) = T_s\,\text{rect}\left(\frac{f}{2f_0}\right),\ B \leq f_0 \leq f_s - B$ | 21-22 4(b) |


Free shared copy · Nur Mohammad

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Delta modulation** | | | | |
| 1 | 20-21 3(a) | 5 | theory | Define slope overload distortion and granular noise in delta modulation and how it is removed in ADM? |
| 2 | 20-21 3(b) | 5 | diagram | Draw the transmitter and receiver circuits block diagram of delta modulation. |
| 3 | 21-22 5(a) | 6 | theory | Explain the principle of sigma-delta modulator. Also explain how it is advantageous over DM. |
| **PCM** | | | | |
| 4 | 20-21 3(c) | 5 | diagram | Explain the basic block diagram of PCM system. |
| 5 | 21-22 4(a) | 5 | theory | What is PCM? Explain PCM system. |
| 6 | 21-22 4(c) | 5 | theory | Write short note on: Companding, Information rate, and Entropy of a source. |
| 7 | 21-22 4(e) | 5 | numerical | An audio signal is sampled at 8 kHz rate using a 32-level quantizer. Find- (i) Peak signal power to peak quantization noise power. (ii) System bandwidth. |
| **Sampling** | | | | |
| 8 | 20-21 4(c) | 7 | derivation | State and prove Nyquist sampling theorem. Also find the interpolation formula for the reconstruction of analog signals from its uniform samples. |
| 9 | 21-22 4(b) | 5 | theory | "A proper valued LPF filter can be used to reconstruct the sampled signal"-justify the statement with the proper mathematical explanations. |

## 4.7. Digital modulation:

43 marks, 7 questions, in 2 of 2 papers. Book: Lathi; Forouzan Ch. 5.

Most often theory (5 of 7). Subtopics: FSK; QAM; DPSK; ASK, FSK and PSK.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| FSK | FSK frequencies | $f_{1,0} = f_c \pm \Delta f$ | 20-21 6(a) |
| | **FSK bandwidth** | $B_{FSK} \approx 2\Delta f + 2R_b$ | 20-21 6(a) |
| | FSK bandwidth (Forouzan) | $B = (1+d)S + 2\Delta f$ | 20-21 6(a), 21-22 4(d) |
| QAM | **QAM signal** | $\varphi_{QAM} = m_1 \cos\omega_c t + m_2 \sin\omega_c t$ | 20-21 6(b) |
| | QAM with phase error | $m_1 \cos\theta - m_2 \sin\theta$ | 20-21 6(b) |
| | **QAM signal** | $\varphi_{QAM}(t) = m_1(t) \cos\omega_c t + m_2(t) \sin\omega_c t$ | 21-22 5(b) |
| | QAM with phase error | $m_1(t) \cos\theta - m_2(t) \sin\theta$ | 21-22 5(b) |
| DPSK | **DPSK encoder** | $q_k = I_k \oplus q_{k-1}$ | 20-21 6(d), 21-22 5(c) |
| | DPSK receiver | $y_k \propto a_k a_{k-1}$ | 20-21 6(d), 21-22 5(c) |
| ASK, FSK and PSK | ASK and PSK bandwidth | $B = (1+d)S$ | 21-22 4(d) |
| | Coherent BASK output | $dA\cos^2\omega_c t \rightarrow \frac{dA}{2}$ | 21-22 5(d) |


Free shared copy · Nur Mohammad

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **FSK** | | | | |
| 1 | 20-21 6(a) | 8 | theory | Briefly describe FSK generation with the necessary block diagram. Include its spectrum and discuss the bandwidth requirements, along with the advantages and disadvantages of FSK. |
| **QAM** | | | | |
| 2 | 20-21 6(b) | 7 | theory | With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks. |
| 3 | 21-22 5(b) | 7 | theory | With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks. |
| **DPSK** | | | | |
| 4 | 20-21 6(d) | 4 | numerical | Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010011100' |
| 5 | 21-22 5(c) | 6 | numerical | Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010010110'. |
| **ASK, FSK and PSK** | | | | |
| 6 | 21-22 4(d) | 5 | theory | Name three mechanisms for the modulation of digital data into an analog signal. Show the differences among them. |
| 7 | 21-22 5(d) | 6 | theory | Briefly describe coherent and non-coherent detection of BASK and BFSK. |

## 4.8. Multiplexing and multiple access:

46 marks, 9 questions, in 2 of 2 papers. Book: Forouzan Ch. 6, 12.

Most often theory (6 of 9). Subtopics: Multiple access; FDM; TDM; Spread spectrum.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| FDM | **FDM link bandwidth** | $B_{link} = NB_{ch} + (N-1)B_g$ | 20-21 2(b) |
| TDM | Quantization step | $\Delta = \frac{2m_p}{L}$ | 20-21 5(c) |
| | Largest error | $e_{max} = \frac{\Delta}{2} = \frac{m_p}{L}$ | 20-21 5(c) |
| | **TDM bit rate** | $R_b = Nf_sn$ | 20-21 5(c), 21-22 6(b) |
| | Minimum bandwidth | $B_T = \frac{R_b}{2}$ | 20-21 5(c) |
| | Levels from an error limit | $\frac{m_p}{L} \leq \epsilon\, m_p \;\Rightarrow\; L \geq \frac{1}{\epsilon}$ | 21-22 6(b) |
| | Synchronous TDM link | $R_{link} = \sum R_i$ | 21-22 6(a) |
| Spread spectrum | FHSS channels | $M = 2^k,\ B_{SS} \approx MB$ | 20-21 6(c) |

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Multiple access** | | | | |
| 1 | 20-21 2(a) | 5 | theory | Differentiate between multiplexing and multiple access technologies. |
| 2 | 20-21 5(a) | 5 | theory | Write down the comparison between TDMA and CDMA. |
| 3 | 21-22 6(c) | 3 | theory | Compare between FDMA and CDMA. |
| 4 | 21-22 6(e) | 3 | theory | Differentiate between multiplexing and multiple access technologies. |
| **FDM** | | | | |
| 5 | 20-21 2(b) | 5 | numerical | Five channels, each with 100 kHz bandwidth, are to be frequency division multiplexing together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channel to prevent interference? |
| **TDM** | | | | |
| 6 | 20-21 5(c) | 7 | numerical | It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patents *[sic: should be "patients"]* are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitude is 0.25% of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data. |
| 7 | 21-22 6(a) | 6 | theory | Write short notes on: Multilevel multiplexing and Multiple slot allocation. |
| 8 | 21-22 6(b) | 6 | numerical | It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patients are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitudes is 0.25 % of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data. |
| **Spread spectrum** | | | | |
| 9 | 20-21 6(c) | 6 | theory | Explain the principle of frequency hopping spread spectrum technique. |

## 4.9. Information theory:

11 marks, 2 questions, in 2 of 2 papers. Book: Lathi.

Most often numerical (2 of 2). Subtopics: Entropy.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Entropy | Information in a symbol | $I = \log_2 \frac{1}{P}$ bits | 20-21 1(a) |
| | **Entropy** | $H = \sum_i P_i \log_2 \frac{1}{P_i}$ bits/symbol | 20-21 1(a) |
| | **Binary entropy** | $H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$ | 20-21 1(a), 21-22 1(a) |
| | Information in a symbol | $I(m_i) = \log_2 \frac{1}{P_i}$ bits | 21-22 1(a) |
| | **Entropy** | $H(m) = \sum_{i=1}^{n} P_i \log_2 \frac{1}{P_i}$ bits/symbol | 21-22 1(a) |


Free shared copy · Nur Mohammad

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| | Bounds of entropy | $0 \leq H(m) \leq \log_2 n$ | 21-22 1(a) |

**Questions:**

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Entropy** | | | | |
| 1 | 20-21 1(a) | 5 | numerical | Find the entropy of single bit binary memoryless source. Also draw the curve of entropy Vs. probability for that source. |
| 2 | 21-22 1(a) | 6 | numerical | Find the entropy $H(m)$ of a memoryless source. |

### 4.10. System design:

7 marks, 1 question, in 1 of 2 papers. Book: Lathi.

Most often theory (1 of 1). Subtopics: Link design.

**Questions:**

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Link design** | | | | |
| 1 | 21-22 6(d) | 7 | theory | Discuss the design procedures of a communication system. |

## 5. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

| No. | Paper | Marks | Derivation | Outline |
|---|---|---|---|---|
| **Signals and spectrum** | | | | |
| 1 | 21-22 1(d) | 6 | Distortionless transmission | Write the output as a delayed scaled input; Fourier transform it; read magnitude and phase conditions. |
| **Amplitude modulation** | | | | |
| 2 | 20-21 4(b) | 7 | VSB | Shape DSB-SC with a filter; multiply by the carrier; filter; require the sum of the shifted responses to be constant. |
| 3 | 21-22 2(d) | 6 | VSB | Write the VSB spectrum; shift by the carrier; keep the baseband terms; require a constant bracket. |
| **Sampling and pulse modulation** | | | | |
| 4 | 20-21 4(c) | 7 | Sampling | Multiply by an impulse train; write its Fourier series; read off the repeated spectrum; filter with an ideal LPF. |

## 6. Repeat tracker:

No repeats among these papers.

## 7. Question style:

How each topic was asked, as a number of questions.

| Topic | Theory | Derivation | Numerical | Diagram |
|---|---|---|---|---|
| Signals and spectrum | 2 | 1 | · | · |
| Noise and channel capacity | 1 | · | 3 | · |
| Amplitude modulation | 4 | 2 | 3 | 1 |
| Angle modulation | 3 | · | 2 | · |
| Receivers | 1 | · | · | · |

| Topic | Theory | Derivation | Numerical | Diagram |
|---|---|---|---|---|
| Sampling and pulse modulation | 5 | 1 | 1 | 2 |
| Digital modulation | 5 | · | 2 | · |
| Multiplexing and multiple access | 6 | · | 3 | · |
| Information theory | · | · | 2 | · |
| System design | 1 | · | · | · |

# 8. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

| No. | Topic | Marks | Covered so far |
|---|---|---|---|
| 1 | Amplitude modulation | 57 | 19 % |
| 2 | Sampling and pulse modulation | 48 | 36 % |
| 3 | Multiplexing and multiple access | 46 | 52 % |
| 4 | Digital modulation | 43 | 66 % |
| 5 | Angle modulation | 29 | 76 % |
| 6 | Noise and channel capacity | 26 | 85 % |
| 7 | Signals and spectrum | 19 | 91 % |
| 8 | Information theory | 11 | 95 % |
| 9 | Receivers | 7 | 98 % |
| 10 | System design | 7 | 100 % |


Free shared copy · Nur Mohammad